

Temperature Regulating System Design for Accelerating Sections 2,3 and 4

CLS DESIGN NOTE 1.2.47.3

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Canadian Light Source
107 North Road
University of Saskatchewan
Saskatoon, Saskatchewan, Canada

	Signature	Date
	<i>Original on File – Signed by:</i>	
Author	_____ Jason Fielden	_____
Reviewer #1	_____ Neil Johnson	_____
Reviewer #2	_____ Dan Lowe	_____
Reviewer #3	_____ Dimo Yosifov	_____
Approver	_____ Mark de Jong	_____

Revision History

Revision	Date	Description	Author
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Table of Contents

1. Introduction..... 1

2. Temperature Regulation System Layout 2

3. System Components 2

4. Pump Sizing..... 2

5. References..... 5

1. Introduction

The Canadian Light Source or CLS is a national facility on the University of Saskatchewan Campus in Saskatoon, Saskatchewan, Canada. The CLS is a 3rd generation synchrotron light source capable of producing high intensity infrared, visible, ultraviolet and x-ray radiation.

Sections 2, 3 and 4 of the CLS linear accelerator or LINAC must be maintained at $40^{\circ}\text{C} \pm 0.1^{\circ}\text{C}$ when operating. During SAL operations, the LINAC was used to operate at a much higher power than will be required for the CLS. As a result, these sections were cooled with three large pumps via three plate to plate heat exchangers. Under the new load regime, the heat exchangers will not be required. In fact, it is anticipated that the water within the systems will actually be cooled during circulation. Therefore, the heaters that exist in the three systems will be used to heat the water. Calculations show that the sections should each receive $3.2 \times 10^{-4} \text{ m}^3/\text{sec.}$ or 5 USGPM of water to achieve proper temperature regulation¹. The system head loss has been calculated at various flow rates to arrive at a head loss versus flow rate curve. This curve has been used in selecting a 7 stage Goulds Booster Pump pump with a $\frac{1}{2}$ horsepower engine for each system.

2. Temperature Regulation System Layout

The skid units for the regulating systems will be located in room 0009 where the current skids are located. Design Note 1.2.47.1² should be consulted for a process flow diagram of the new systems.

The majority of the components and piping for the new temperature regulation loops can be reused from the existing systems. The piping will be changed as little as possible while placing the new pumps. Gauging has been placed periodically to monitor performance. In addition, ball valves have been employed in several locations to allow for isolation of components. The three way valve and heat exchanger will remain in the system for a period to ensure that they will not be required due to an overheating situation.

3. System Components

The section 2, 3 and 4 temperature regulation loops will be composed of drawn copper piping and tubing. The sizes will be in accordance with typical sizes used at the CLS. These sizes are $\frac{3}{8}$ " tube, $\frac{1}{2}$ " pipe, $\frac{3}{4}$ " pipe, 1" pipe, 1½" pipe, 2" pipe and 2½" pipe. Emco pipe, tube, valves and fittings are presently used at the CLS. Therefore, any new equipment must be compatible.

The system will require some new diagnostic equipment. Design note 1.2.47.1 shows the specifications for these components².

4. Pump Sizing

The sizing of the pump depends on the size and routing of the piping through which the cooling water is pumped. Calculations show that a flow of $3.2 \times 10^{-4} \text{ m}^3/\text{sec.}$ or 5 USGPM is required. A plot of flow versus pressure drop must be determined to size a suitable pump.

The head loss of the system will be defined by the head loss from the pump to the section plus the head loss through the section, plus the head loss from the section back to the pump. To establish a system head loss curve, the cooling loop was tested with the current pump and the three way valve directing water away from the heat exchanger. This setup closely resembles the new system. A ball valve was used to restrict flow to a number of different values. These flow values were read off of the pump curve of the existing pump at the pressure differential as shown on the pressure transducers in the system. The head loss across the system was observed at all of these flow rates. In this way, a curve of flow versus head loss was established. Finally, this curve was extrapolated back to zero flow. Figure 4.1 shows this curve.

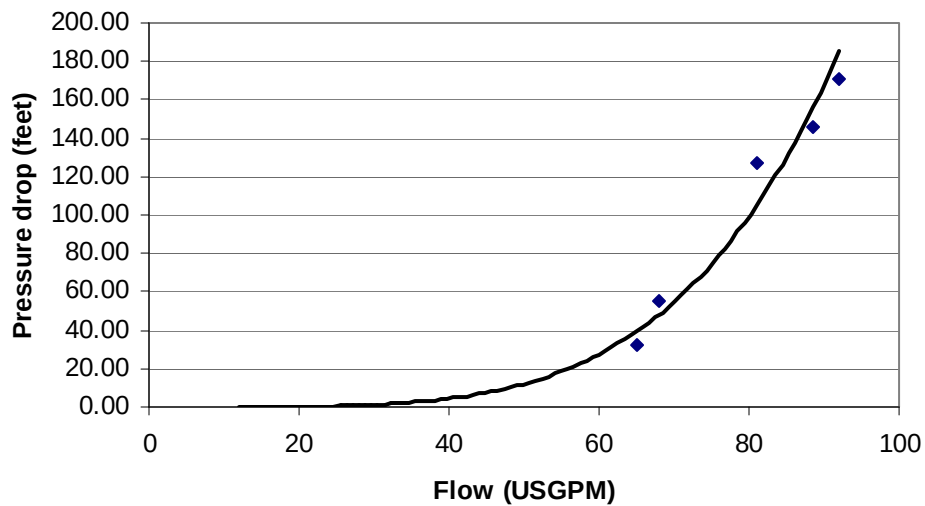


Figure 4.1: Pressure drop versus flow curve for section 2, 3, 4 temperature regulation loops

A pump was sized to provide the desired flow while considering the system head loss curve of Figure 4.1. A 7 stage Goulds Booster Pump with a $\frac{1}{2}$ horsepower motor that is compatible with an ultra-pure de-ionized water system will be implemented. The performance curve for the Goulds pump is shown in Figure 4.2.

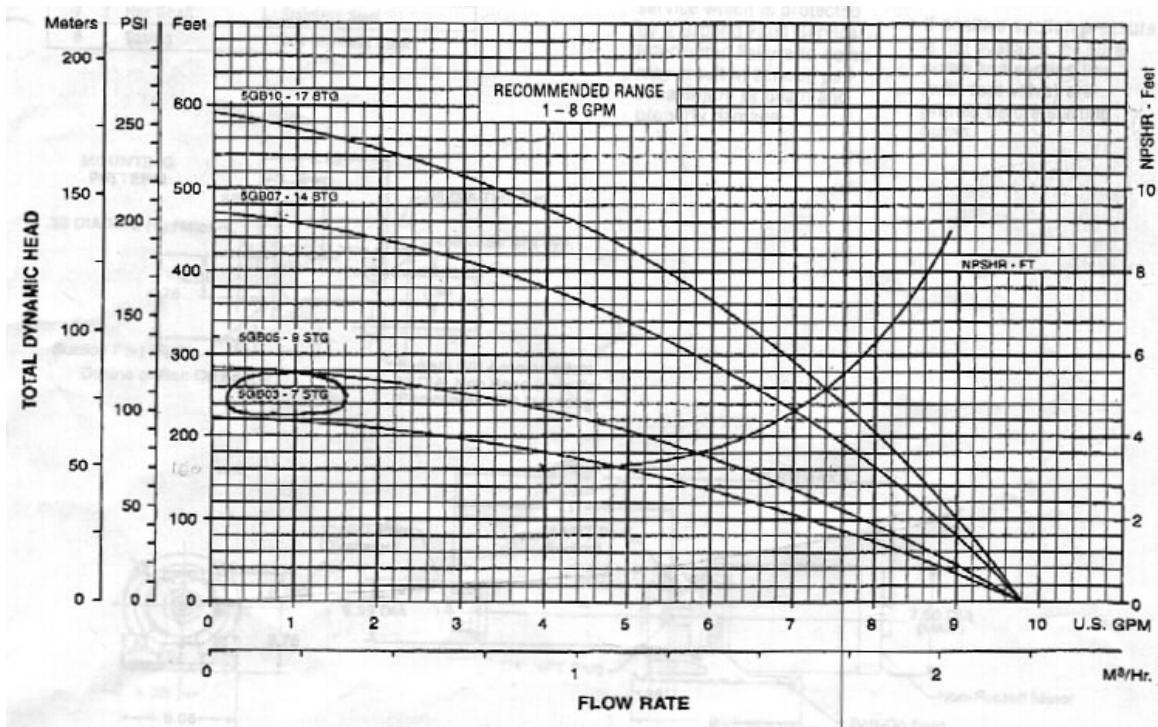


Figure 4.2: Performance curve for 7 stage Goulds Booster Pump

When comparing the system curve with the pump curve, it can be seen that there is very little resistance for the pump to work on in the flow range of interest. As a result, it is recommended that the ball valve on the return side of the section be partially closed to give some resistance. This valve should be closed to reach a minimum pressure drop across the system of 207 000 Pa or 30 psi which corresponds to a flow rate of 5.05×10^{-4} m³/s or 8 GPM. At this operating condition, the pump is heating the water with approximately 120 W.

5. References

1. Calculations performed by Dr. Jake Haimson
2. “Section Heat Exchanger Upgrade”, Johnson, Neil, (Canadian Light Source, Saskatoon, Saskatchewan, 2000).